

Capstone Checkpoint 6: Safety, Guardrails, and Human Intervention Plan

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1. Intended Scope and Primary Safety Risks

The **mail-organizer-pro** agentic local utility is designed to autonomously ingest, parse, classify, and archive physical and digital document streams (such as bills, tax forms, and legal correspondence). As established in previous checkpoints (specifically Checkpoint 1, 2, and 5), this pipeline uses a local, privacy-first 2-agent architecture where an Extraction Agent maps unstructured documents to intermediate schemas and a Verification Agent executes database commits and directory lookups. While this structure tames noisy physical inputs, expanding its autonomy introduces three critical system-level safety risks:

- **Taxonomy Fragmentation & Synonym Hallucinations:** Standard language models permitted to generate directory paths dynamically tend to hallucinate folder synonyms (e.g., routing utility bills to 'Home/Power' during one run and 'Utilities/Electric' in the next), rapidly corrupting local filesystem structures.
- **Destructive Filesystem Actions:** Unconstrained file write or move privileges can lead to accidental directory overwrites, file duplication, or systemic data loss of unbacked physical scans when duplicate names conflict.
- **Retrieval and Contextual Drift:** When executing few-shot retrieval via local vector storage, the system risks injecting false-positive contexts (e.g., matching a Duquesne Light power bill with a PGW gas bill layout due to highly overlapping template structures and/or entity descriptions, e.g.; Utility company), resulting in silent misfiling and poor meta data field resolutions.

2. Core Multi-Point Agent Guardrails

To mitigate these risks, the pipeline implements a multi-layered guardrail strategy integrated directly into the 2-agent coordination lifecycle:

- **Pre-Generation Constraints:** A SHA-256 file hashing module enforces instant local deduplication to prevent redundant storage processing. OCR ingestion pipelines strictly slice and limit the extracted text buffer to the first and last 1,000 characters to isolate identity markers (e.g., account numbers) and restrict token inflation.
- **During-Generation Constraints:** The Extraction Agent's generative output is strictly bound to structured intermediate Pydantic JSON schemas. Schema validation enforces type-checking on key metadata (dates, vendors, dollar amounts) and requires the LLM to output a calibrated field-level confidence score.
- **Post-Generation Enforcement:** An Anchor-Key Validation Rule enforces strict alphanumeric matching (such as matching specific account numbers) before RAG context is injected. Crucially, the Verification Agent resolves directory paths via a 100% deterministic local mapping table. LLMs are never permitted to generate directories; routing remains an immutable lookup.

3. Evaluation and Observability Metrics

To monitor the agent's safe and effective operation in production, the system evaluates performance across four primary axes:

- **Metadata Extraction Correctness:** Measured via exact match accuracy of extracted fields (Vendor, Date, Total) against a ground-truth dataset compiled from past manual reviews.
- **Calibration & Expected Calibration Error (ECE):** Measures the alignment between the agent's self-assessed confidence scores and its empirical accuracy (e.g., an 85% confidence score should yield correct extraction 85% of the time) to actively prevent overconfidence-driven hallucinations.
- **RAG Groundedness:** Ensures that any injected context is strictly sourced from verified, historical manual corrections without outside knowledge leakage.
- **Escalation and Fallback Success Rate:** Tracks the percentage of low-confidence or failed validation runs that are safely diverted to the manual review queue without causing execution failures or system-state corruption.

4. Selective Human Intervention and Feedback Loop

Rather than relying on universal, manual oversight that destroys automation velocity, the Verification Agent acts as a local circuit breaker, routing documents to a local human-in-the-loop review queue under three strict trigger conditions: (1) if the Extraction Agent's field-level confidence score drops below 0.85, (2) if Pydantic schema validation fails to parse the core required fields, or (3) if the Anchor-Key validation rules fail to verify a retrieved vector match.

The human review interface is designed for high-velocity resolution, presenting focused contextual metadata alongside the original document scan. This manual intervention feeds a critical, self-correcting feedback loop: when a user manually overrides or corrects a record, a Structured Correction Payload is generated, paired with the 1,000-character document boundary slice and a human readable summary of the document as well as a record of agent actions, embedded locally, and committed back to the relational SQLite database and vector index. In future runs, the agent queries these verified past corrections as grounded few-shot examples, turning past human intervention directly into autonomous system improvement. In the event of continued low-confidence scores it can re-enter the manual review bucket.

5. Strategic Coordination and Architectural Trade-offs

Safety Coordination Overview: The guardrails, evaluation metrics, and human review systems operate as a closed-loop defense in depth. Pre-generation filters block bad data at ingestion, and guide the LLM; during-generation schema boundaries enforce structural compliance; post-generation validation catches logical drift; and selective escalation isolates unresolvable exceptions while routing their manual corrections back into the vector index as grounding memory.

Autonomy vs. Reliability Trade-off: In establishing this control system, generative autonomy was deliberately sacrificed to achieve absolute storage reliability. While allowing the LLM to dynamically create paths and manage folders would increase autonomous flexibility, the resulting risk of taxonomy fragmentation made deterministic mapping tables a superior architectural choice. Standardizing directory paths as immutable

Constants ensures 100% filesystem integrity. Perhaps there is an opportunity to involve a multi-tier sub system to automate the directory structure path generation and recall, the scope of this project as reduced as to not re-litigate concepts being demonstrated in other parts of the system.

Efficiency vs. Overhead Trade-off: As detailed in Checkpoint 4.1, complex branching architectures like Tree-of-Thought (ToT) were dismissed due to the prohibitive computational overhead they impose on local consumer hardware. Instead, we pair a single-stream Constrained ReAct loop with Pydantic validation and local RAG retrieval. This delivers sub-second, highly predictable execution with a minimal local hardware footprint.